



Input

Target Schema

Table-level Specification

I need a table containing ratings by genre for each director in 2019 and 2020.

Column-level Specification

director_name: The name of the director
genres: A single movie genre category.
2019: Average rating for movies in 2019.
2020: Average rating for movies in 2020.

Source Tables

Table: movies

id	title	dir_id	genres
1	jurassic park	101	Sci-Fi
1	jurassic park	102	Sci-Fi
...	...	...	...
17	the dark knight	103	Sci-Fi, Action

Table: ratings

movie	values	rating
Jurassic...	8.2 (2022)	8.2
...	...	...

Table: directors

id	?
101	Steven ...
...	...

(a) Code Generation

```

...
merged =
movies.merge(director
s,
left_on='dir_id',righ
t_on='id').merge(rati
ngs,left_on='title',r
ight_on='movie')
merged[['score',
'year']] =
merged['rating'] ...

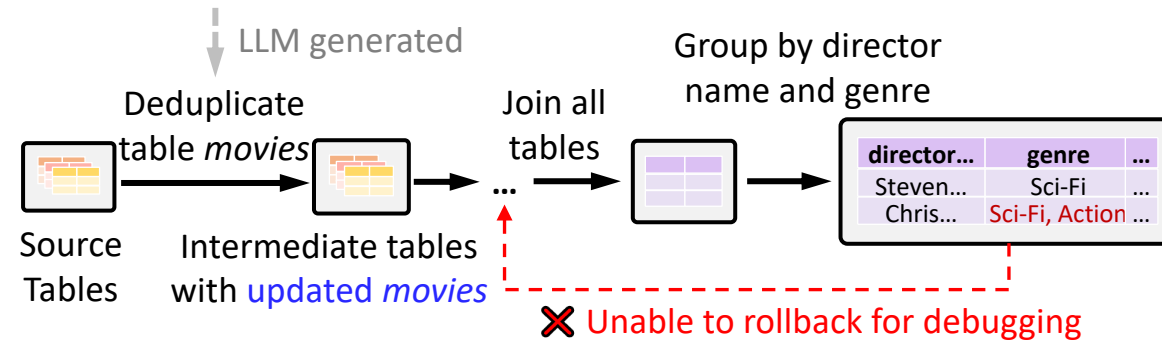
```

LLM

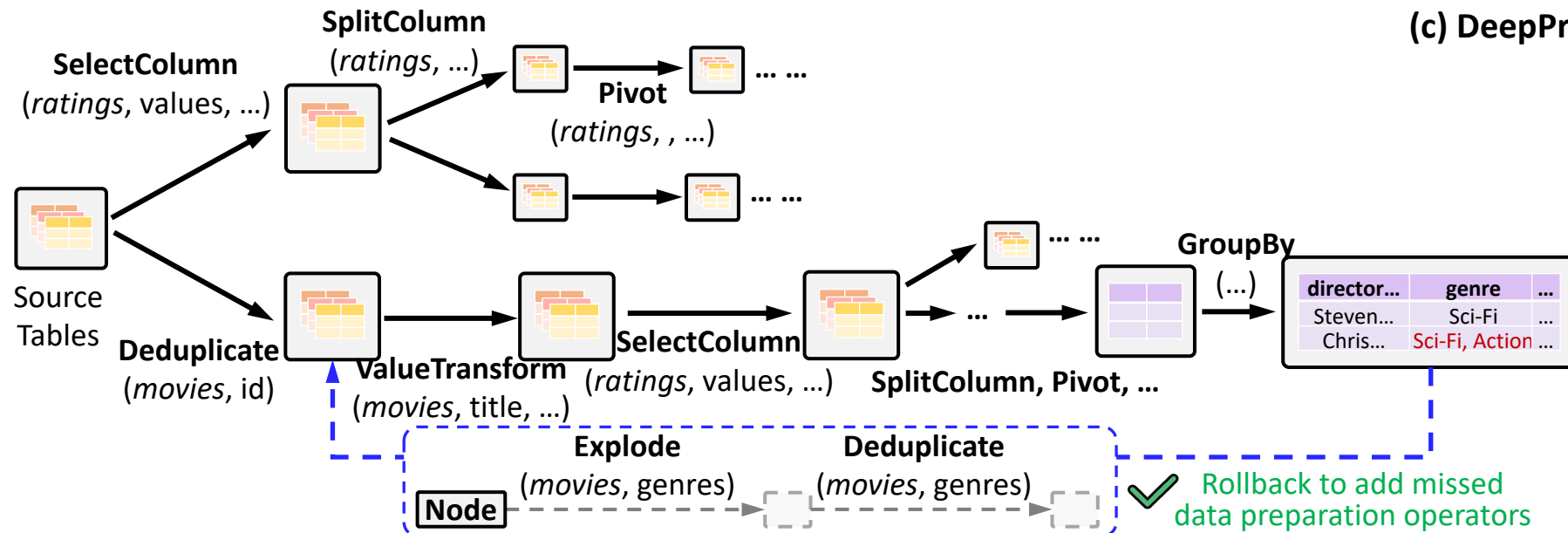
The Result
Table is
Empty

Executor

(b) ReAct



(c) DeepPrep



Wrong Target Table

director_name	genres	2019	...
Steven Spielberg	Sci-Fi	7.6	...
James Cameron	Sci-Fi	8.9	...
Christopher Nolan	Sci-Fi, Action	8.95	...
...	...	...	...

Correct Target Table

director_name	genres	2019	2020
Steven Spielberg	Sci-Fi	7.6	7.7
James Cameron	Sci-Fi	8.9	9.2
Christopher Nolan	Action	8.95	8.8
Christopher Nolan	Sci-Fi	8.9	8.8
...	...	...	...